

# Haocheng Xi

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## EDUCATION

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### University of California, Berkeley

Ph.D. in Computer Science, Berkeley AI Research (BAIR)

Advisor: [Prof. Kurt Keutzer](#)

Berkeley, CA

09/2024 – Present

### Tsinghua University

B.Eng. in Computer Science & Technology

Institute for Interdisciplinary Information Sciences (IIIS)

Yao Class, led by [Prof. Andrew C.C. Yao](#)

Beijing, China

09/2020 – 06/2024

### University of Washington

Visiting Student, Paul G. Allen School of Computer Science & Engineering

Advisor: [Prof. Sheng Wang](#)

Seattle, WA

02/2023 – 08/2023

### Beijing No.8 High School

[Experimental class](#) for gifted and talented young, Excellent Graduate

Beijing, China

09/2015 – 07/2020

## RESEARCH INTERESTS

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My research interests lie in efficient machine learning, such as quantization and sparsity. I aim to push the boundaries of how we can effectively compress and accelerate deep learning models while maintaining their accuracy and robustness.

## SELECTED PUBLICATIONS

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### *Efficient Video Generation*

#### Quant VideoGen: Auto-Regressive Long Video Generation via 2-Bit KV-Cache Quantization

[Haocheng Xi\\*](#), [Shuo Yang\\*](#), [Yilong Zhao](#), [Muyang Li](#), [Han Cai](#), [Xingyang Li](#), [Yujun Lin](#), [Zhuoyang Zhang](#), [Jintao Zhang](#), [Xiuyu Li](#), [Zhiying Xu†](#), [Jun Wu†](#), [Chenfeng Xu](#), [Ion Stoica](#), [Song Han](#), [Kurt Keutzer](#)  
International Conference on Machine Learning (ICML), 2026. [\[arxiv\]](#) [\[code\]](#)

#### Sparse VideoGen: Accelerating Video Diffusion Transformers with Spatial-Temporal Sparsity

[Haocheng Xi\\*](#), [Shuo Yang\\*](#), [Yilong Zhao](#), [Chenfeng Xu](#), [Muyang Li](#), [Xiuyu Li](#), [Yujun Lin](#), [Han Cai](#), [Jintao Zhang](#), [Dacheng Li](#), [Jianfei Chen](#), [Ion Stoica](#), [Kurt Keutzer](#), [Song Han](#)  
International Conference on Machine Learning (ICML), 2025. [\[arxiv\]](#) [\[code\]](#) [\[website\]](#) [\[poster\]](#)

#### Sparse VideoGen 2: Accelerate Video Generation with Sparse Attention via Semantic-Aware Permutation

[Shuo Yang\\*](#), [Haocheng Xi\\*](#), [Yilong Zhao](#), [Muyang Li](#), [Jintao Zhang](#), [Han Cai](#), [Yujun Lin](#), [Xiuyu Li](#), [Chenfeng Xu](#), [Kelly Peng](#), [Jianfei Chen](#), [Song Han](#), [Kurt Keutzer](#), [Ion Stoica](#)  
Conference on Neural Information Processing Systems (NeurIPS), 2025. [\[arxiv\]](#) [\[code\]](#) [\[website\]](#)  
Selected as **Spotlight Paper** in NeurIPS 2025.

#### Flash-KMeans: Fast and Memory-Efficient Exact K-Means

[Shuo Yang\\*](#), [Haocheng Xi\\*](#), [Yilong Zhao](#), [Muyang Li](#), [Xiaoze Fan](#), [Jintao Zhang](#), [Han Cai](#), [Yujun Lin](#), [Xiuyu Li](#), [Kurt Keutzer](#), [Song Han](#), [Chenfeng Xu](#), [Ion Stoica](#)  
Arxiv, 2026. [\[arxiv\]](#) [\[code\]](#)

### **Radial Attention: $O(n \log n)$ Sparse Attention with Energy Decay for Long Video Generation**

Xinyang Li\*, Muyang Li\*, Tianle Cai, Haocheng Xi, Shuo Yang, Yujun Lin, Lvmin Zhang, Songlin Yang, Jinbo Hu, Kelly Peng, Maneesh Agrawala, Ion Stoica, Kurt Keutzer, Song Han  
Conference on Neural Information Processing Systems (NeurIPS), 2025. [\[arxiv\]](#) [\[code\]](#) [\[website\]](#)

### **StreamDiffusionV2: A Streaming System for Dynamic and Interactive Video Generation**

Tianrui Feng, Zhi Li, Shuo Yang, Haocheng Xi, Muyang Li, Xiuyu Li, Lvmin Zhang, Keting Yang, Kelly Peng, Song Han, Maneesh Agrawala, Kurt Keutzer, Akio Kodaira, Chenfeng Xu  
Conference on Machine Learning and Systems (MLSys), 2026. [\[arxiv\]](#) [\[code\]](#) [\[website\]](#)  
Selected as **Best Paper** in MLSys 2026.

## ***Efficient Language Model***

### **LoSA: Locality Aware Sparse Attention for Block-Wise Diffusion Language Models**

Haocheng Xi, Harman Singh, Yuezhou Hu, Coleman Hooper, Rishabh Tiwari, Aditya Tomar, Minjae Lee, Wonjun Kang, Michael Mahoney, Chenfeng Xu, Kurt Keutzer, Amir Gholami  
International Conference on Machine Learning (ICML), 2026. [\[arxiv\]](#)

### **COAT: Compressing Optimizer states and Activation for Memory-Efficient FP8 Training**

Haocheng Xi, Han Cai, Ligeng Zhu, Yao Lu, Kurt Keutzer, Jianfei Chen, Song Han  
International Conference on Learning Representations (ICLR), 2025. [\[arxiv\]](#) [\[code\]](#) [\[website\]](#)

### **Jetfire: Efficient and Accurate Transformer Pretraining with INT8 Data Flow and Per-Block Quantization**

Haocheng Xi, Yuxiang Chen, Kang Zhao, Kai Jun Teh, Jianfei Chen, Jun Zhu  
International Conference on Machine Learning (ICML), 2024. [\[arxiv\]](#) [\[code\]](#) [\[poster\]](#)  
Selected as **Spotlight Paper** in ICML 2024.

### **Training Transformers with 4-bit Integers**

Haocheng Xi, Changhao Li, Jianfei Chen, Jun Zhu  
Conference on Neural Information Processing Systems (NeurIPS), 2023. [\[arxiv\]](#) [\[code\]](#)

### **QuantSpec: Self-Speculative Decoding with Hierarchical Quantized KV Cache**

Rishabh Tiwari\*, Haocheng Xi\*, Aditya Tomar, Coleman Hooper, Sehoon Kim, Maxwell Horton, Mahyar Najibi, Michael W. Mahoney, Kurt Keutzer, Amir Gholami  
International Conference on Machine Learning (ICML), 2025. [\[arxiv\]](#) [\[poster\]](#)

### **Oscillation-Reduced MXFP4 Training for Vision Transformers**

Yuxiang Chen, Haocheng Xi, Jun Zhu, Jianfei Chen  
International Conference on Machine Learning (ICML), 2025. [\[arxiv\]](#)

## **INTERNSHIP EXPERIENCE**

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Nvidia Research, Research Intern

03/2024 – 08/2024, 2025/05 – Present

Advisor: [Prof. Song Han](#)

### **Jet-RL: On-Policy FP8 Reinforcement Learning**

- Proposed Jet-RL, an FP8 reinforcement learning framework that aligns training and rollout precision to reduce numerical policy mismatch.
- Identified instability in BF16-train and FP8-rollout pipelines under long-rollout generation and challenging reasoning tasks.
- Achieved up to 33% rollout speedup, 41% training speedup, and 16% end-to-end speedup while preserving stable convergence.

### **COAT: Compressing Optimizer states and Activation for Memory-Efficient FP8 Training**

- Introduced COAT, a framework that quantizes optimizer states and activations to FP8 precision, significantly reducing memory usage during large-scale model training.
- Proposed Dynamic Range Expansion for Optimizer states and Mixed-Granularity Activation Quantization, achieving outstanding accuracy and efficiency.
- Achieved a  $1.54\times$  reduction in training memory footprint and a  $1.43\times$  speedup compared to BF16 training, also doubled the training batch size to utilize GPU better.
- Training loss curve and downstream task performance were consistent with BF16 training, across language models and vision language models.

## HONORS

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**Fellowship of Tsinghua Xuetang Talents Program** Among top 300 / 3000 Tsinghua students each year

**Athletic Excellence Scholarship** In 2022

**First Prize of National Senior High School Mathematics Competition** In 2019

## SKILLS

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**Language:** TOFEL: Total 110 (Reading 29, Listening 29, Speaking 24, Writing 28)

GRE: Quantitative 170, Verbal 158, Writing 4.0

**Programming and Software:** Python, CUDA, C++, Bash, Git,  $\text{\LaTeX}$

**Deep Learning Package:** PyTorch, Transformers, Triton, PEFT, TransformerEngine, VeRL

**Conference review:** ICML (2025), NeurIPS (2024, 2025), ICLR (2025, 2026)